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Roll No

IT-8003 (2) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Data Science
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

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| 1. a) Explain big data management architecture with neat block diagram. | 7 |
| b) Explain the role of relational databases in big data. | 7 |
| 2. a) Classify the different data types with reference to big data. | 7 |
| b) Define a normal distribution and derive the mean, median, mode of a normal distribution. | 7 |
| 3. a) Derive the mean and the variance of a binomial distribution. | 7 |
| b) Explain the role of support vector machine in data analysis. | 7 |
| 4. Explain the following: | 14 |
| i) Naive bayes classifier | |
| ii) Linear regression with nonlinear substitution | |
| iii) Evolution of data management | |
| 5. a) Explain parallel vs distributed processing. | 7 |
| b) Explain data warehousing architecture. | 7 |
| 6. a) Explain data preparation, model planning and model building. | 7 |
| b) Differentiate cluster architecture and traditional architecture. | 7 |
| 7. a) Explain core components of hadoop ecosystem with neat diagram. | 7 |
| b) Explain loops and conditional execution in R language. | 7 |
| 8. a) Explain data manipulation and statistical analysis with R. | 7 |
| b) Explain various data structures used in R. | 7 |
